

A GUIDE TO TYPES OF SHIPS

Study Notes — Container • Bulk • Tanker • Passenger
Offshore • Fishing • Speciality • High-Speed • Dredgers

OVERVIEW

Cargo ships are classified by purpose, size, and type of cargo carried. The economic factor is central to merchant ship design — owners want maximum return on investment, so design depends on both current economic needs and future adaptability.

From a vessel's preliminary design, the following can be obtained: dimensions, displacement, stability, propulsive characteristics & hull form, preliminary general arrangement, and principal structural details. The type of ship determines these parameters.

Main Topics at a Glance

- Container Ships
- Bulk Carrier
- Tanker Ships
- Roll-on Roll-Off (Ro-Ro) Ships
- Passenger Ships
- Offshore Vessels
- Fishing Vessels
- Speciality Vessels
- High-Speed Craft
- Dredgers

1 Container Ships

Vessels structured to hold large quantities of cargo packed into standardized containers.

Classified by Size

- Panamax
- Suezmax
- Post-Panamax
- Post-Suezmax
- Post-Malaccamax
- **Refrigerated Container Ships** — carry refrigerated cargo, mainly in reefer containers.

2 Bulk Carrier Ships

Transport dry cargo in bulk (loose, unpackaged) quantities — e.g. food grains, ores, coal, cement.

Types

- Conventional bulkers
- Geared bulker
- Gearless bulker
- Self-discharging bulker
- Lakers
- BIBO

Other Dry-Cargo Forms

- **Tramps** — no fixed schedule or published ports of call.
- **Cargo Liners** — ocean liners on fixed point-to-point routes (e.g. transatlantic Europe–America).

3 Tanker Ships

Specialised vessels carrying large amounts of liquid cargo.

Main Types

- **Oil Tankers** — carry crude oil and by-products.
- **Liquefied Gas Carriers** — transport LPG, LNG, or liquefied chemical gases in bulk.
- **Chemical & Product Carriers** — transport chemicals and liquid products in bulk.
- Others — juice tankers, wine tankers, integrated tug barges.

Classified by Size

- VLCC
- ULCC
- Panamax
- Aframax
- Suezmax
- Capesize
- Handymax
- Lighters
- Handy

4 Roll-on Roll-Off (Ro-Ro) Ships

Ro-Ro (Roll-on/roll-off) vessels carry wheeled cargo.

- Pure Car Carrier (PCC) / Pure Car and Truck Carrier (PCTC)
- ConRo — Container Vessel + Ro-Ro
- GenRo — General Cargo + Ro-Ro
- RoPax
- Complete RoRo Ships

5 Passenger Ships

Mainly used to transit passengers.

Classified Into

- **Ferries** — short-distance passenger & vehicle transit.
- **Cruise Ships** — recreational, floating hotels with luxury facilities.

Further Classified As

- Liners, Cruise Ships, Pilgrimage Ships
- Cross Channel Ferries, Coastal Ferries, Harbour Ferries
- Arctic and Antarctic Cruises

6 Offshore Vessels

Support oil exploration and offshore construction.

Main Types

- Supply Ship — supplies offshore rigs
- Pipe Layers — lay pipes and cables

- Crane Barges / floating cranes — lift heavy loads
- Semi-submersible Drill Rigs — mobile offshore drilling units
- Drill Ships — exploratory offshore drilling / scientific drilling
- Accommodation Barges — floating hotel and/or cargo space
- Production Platforms — extract/process oil & gas, temporary storage
- Floating Storage Unit (FSU) — storage of oil and by-products
- FPSO — floating production, storage and offloading unit
- Anchor Handling Vessels — offshore construction/installation
- Diving Vessels — support divers for underwater jobs

7 Fishing Vessels

Used for recreational or commercial fishing at sea.

- **Trawlers, Purse Seiners** — trawlers drag a net (trawl) through water; purse seiners deploy a wall of netting around a school of fish.
- **Factory Ships** — large ocean-going vessels with on-board processing/freezing facilities for fish or whales.

8 Speciality Vessels

Built and used for specific purposes.

- Tugs — manoeuvre vessels by pushing/towing
- Tenders — support other boats/ships, transport people or supplies
- Pilot Crafts — transport harbour pilots
- Cable Layers — lay cables on the sea bed
- Research Vessels — seismic, hydrographic, oceanographic, polar vessels, etc.
- Salvage Vessels — recovery of lost property at sea
- Lightships — act as a lighthouse where one can't be built
- Barge Carriers — flat-bottomed, for river/canal transport of heavy goods
- Timber Carriers — carry timber
- Livestock Carriers — carry livestock/animals
- Ice Breaker Ships — cut ice deposits to keep waters navigable

9 High-Speed Craft

Technologically advanced, typically high-speed marine vehicles; few technologies used in commercial vessels.

- Multihulls, including wave piercers
- Small Waterplane Area Twin-Hull (SWATH)
- Surface Effect Ship (SES) and Hovercraft
- Hydrofoil
- Wing in Ground Craft (WIG)

10 Dredgers

Used for dredging — an underwater excavation activity in shallow seas/freshwater to gather bottom sediments and widen channels. Used to make coastal areas navigable, for deep-sea mining, etc.

- Mechanical dredgers
- Hydraulic dredgers

QUICK REFERENCE TABLE

Category	Key Sub-types (examples)
Container Ships	Panamax, Suezmax, Post-Panamax, Post-Malaccamax, Reefer containers
Bulk Carriers	Conventional, Geared, Gearless, Self-discharging, Lakers, BIBO
Tanker Ships	Oil, LNG/LPG, Chemical; VLCC, ULCC, Aframax, Suezmax, Capesize
Ro-Ro Ships	PCC/PCTC, ConRo, GenRo, RoPax
Passenger Ships	Ferries, Cruise Ships, Liners, Pilgrimage Ships
Offshore Vessels	Supply Ships, Pipe Layers, Drill Ships, FPSO, FSU, AHVs
Fishing Vessels	Trawlers, Purse Seiners, Factory Ships
Speciality Vessels	Tugs, Tenders, Cable Layers, Research Vessels, Ice Breakers
High-Speed Craft	SWATH, SES/Hovercraft, Hydrofoil, WIG
Dredgers	Mechanical, Hydraulic